

Weakly Stochastic Matrices

Adjacency and Distance Matrices of Trees

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Abstract—This note investigates weakly column- (or row-) stochastic matrices. These are matrices in which certain columns can be selected such that their sum gives the all one column vector. Similarly for rows. A theoretical method is presented that enables one to check whether a matrix is weakly column stochastic by using the row difference matrix. For simple graphs row or column can be omitted as the adjacency matrices and the distance matrices are symmetric. Then another more practical criterion is also presented. Most applications given here are on tree related matrices. We determine which trees with a small number of vertices are weakly stochastic. Results for distance matrices are also presented.

Index Terms—Regular graphs, stochastic matrices, row and column space

I. DEFINITIONS AND FIRST RESULTS

A. Definitions

We define a matrix over a field $\mathbb{F}$ as *weakly column stochastic* if it contains the all-one vector $u = [1, 1, \dots, 1]^T$ in the column space of that matrix. A matrix is *weakly row stochastic* if it contains the all-one vector $u = [1, 1, \dots, 1]$ in the row space of that matrix.

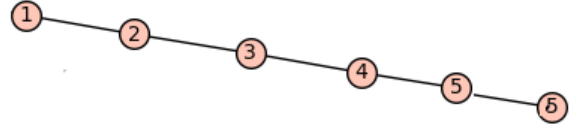
A matrix can be called *quasi column stochastic* if the sum of all columns is a non zero multiple of the all-one vector $u = [1, 1, \dots, 1]^T$. Similarly a matrix can be called *quasi row stochastic* if the sum of all its rows is equal to a multiple of the all-one vector $u = [1, 1, \dots, 1]$ in the row space of that matrix.

The *adjacency matrix* of any simple connected graph $G = (V, E)$ with $|V| = n$ is a $n \times n$ matrix, we denote the vertices of G by $v_1, v_2, \dots, v_n$. Then the adjacency matrix of G is the matrix $A(G) = [a_{ij}]$, where a_{ij} is the number of edges joining v_i and v_j . [1]

B. The difference matrix of a matrix

Let A be any $n \times n$ matrix. We define the row difference matrix of A , $RDf(A)$, as the $\binom{n}{2} \times n$ matrix of all the difference between the rows of the original matrix. This implies that the row space of $RDf(A)$ is $Linspan(a_i - a_j)_{1 \leq i < j \leq n}$ where a_i and a_j are rows of the matrix.

Example 1: For example, let's take the path graph of 6 vertices, and consider its adjacency matrix



$$A(P_6) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \end{matrix}$$

Then the row difference matrix of P_6 is the following 15×6 matrix

$$RDf(P_6) = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ -1 & 1 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & -1 & 1 & 0 & 1 \\ -1 & 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & -1 & -0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{bmatrix} \begin{matrix} a_2 - a_1 \\ a_3 - a_1 \\ a_4 - a_1 \\ a_5 - a_1 \\ a_6 - a_1 \\ a_3 - a_2 \\ a_4 - a_2 \\ a_5 - a_2 \\ a_6 - a_2 \\ a_4 - a_3 \\ a_5 - a_3 \\ a_6 - a_3 \\ a_5 - a_4 \\ a_6 - a_4 \\ a_6 - a_5 \end{matrix}$$

We may consider the part of this matrix that is obtained by taking differences involving only a fixed row a_i , e.g. for the differences with the first row a_1 we get

$$\begin{bmatrix} 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} a_2 - a_1 \\ a_3 - a_1 \\ a_4 - a_1 \\ a_5 - a_1 \\ a_6 - a_1 \end{matrix}$$

Therefore we get

$$RDf_1(P_6) = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Note that RDf_1 is a part of the matrix $RDf(A)$. The same construction may be done by subtracting other rows, for example, in the case of P_6 and the second vector, $RDf_2(P_6)$ is

$$RDf_2(P_6) = \begin{bmatrix} -1 & 1 & -1 & 0 & 0 & 0 & a_1 - a_2 \\ -1 & 1 & -1 & 1 & 0 & 0 & a_3 - a_2 \\ -1 & 0 & 0 & 0 & 1 & 0 & a_4 - a_2 \\ -1 & 0 & -1 & 1 & 0 & 1 & a_5 - a_2 \\ -1 & 0 & -1 & 0 & 1 & 0 & a_6 - a_2 \end{bmatrix}$$

Similarly for $RDf_3(P_6)$ and so on. These smaller matrices are all part of $RDf(P_6)$. But indeed it is easy to see that the linear of the rows of each of the smaller matrices is equal to the linear span of the rows of $RDf(P_6)$.

The same argument also applies to a general matrix A with its row difference matrices $RDf(A)$ and $RDf_k(A)$.

We can note that each row of $RDf_1(A)$ is a linear combination of the rows (indeed a subset of the rows) of $RDf(A)$, thus it suffices to show just $RDf_1(A)$ generates each row of $RDf(A)$:

$$a_i - a_j = (a_i - a_1) - (a_j - a_1).$$

Lemma 1: Assume that A is a $n \times n$ matrix with entries in $\mathbb{F}$.

$$\text{rank}(RDf_1(A)) \in \{\text{rank}(A), \text{rank}(A) - 1\}$$

For the proof let $A = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$ where w_i are rows of the matrix.

Let $r = \text{rank}(A)$, and we may assume that $w_1, w_2, \dots, w_r$ are linearly independent. Then we know by the definition that

$$RDf_1(A) = \begin{bmatrix} w_2 - w_1 \\ w_3 - w_1 \\ \vdots \\ w_r - w_1 \\ \vdots \\ w_n - w_1 \end{bmatrix}$$

We show that the row differences $w_2 - w_1, w_3 - w_1, \dots, w_r - w_1$ are linearly independent.

Assume then that $\sum_{i=2}^r b_i(w_i - w_1) = 0$, so that $\sum_{i=2}^r b_i w_i - (\sum_{i=2}^r b_i) w_1 = 0$. From the linear independence of $w_1, w_2, \dots, w_r$ it follows that $b_2 = b_3 = \dots = b_r = 0$.

Therefore $w_2 - w_1, w_3 - w_1, \dots, w_r - w_1$, are linearly independent.

Hence, $\text{rank}(RDf_1(A)) \geq r - 1$.

We know that $\text{rank}(RDf_1(A)) \leq \text{rank}(A)$. Thus it follows that

$$\text{rank}(RDf_1(A)) \in \{\text{rank}(A), \text{rank}(A) - 1\}.$$

Theorem 1: A matrix A is weakly stochastic if and only if the second alternative of the lemma is valid, i.e.

$$\text{rank}(RDf(A)) = \text{rank}(A) - 1. \quad (1)$$

For the proof assume that A is *weakly column-stochastic*.

Let $A = [c_1, c_2, \dots, c_n]$ where c_i are columns of the matrix.

Then, by the definition of *weakly stochastic*,

$$\exists x = [x_1, x_2, \dots, x_n]^T \neq [0, 0, \dots, 0]^T \in \mathbb{F}^n,$$

such that $u = \sum_{i=1}^n x_i c_i$
where $u = [1, 1, \dots, 1]^T$

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

Then by the definition of weakly stochastic we get the following

$$\begin{aligned} 1 &= \sum_{i=1}^n x_i a_{1i} \\ 1 &= \sum_{i=1}^n x_i a_{2i} \\ &\vdots \\ 1 &= \sum_{i=1}^n x_i a_{ni} \end{aligned}$$

By taking differences of the above equations

$$\begin{aligned} 0 &= \sum_{i=1}^n x_i (a_{2i} - a_{1i}) \\ 0 &= \sum_{i=1}^n x_i (a_{3i} - a_{1i}) \\ &\vdots \\ 0 &= \sum_{i=1}^n x_i (a_{ni} - a_{1i}) \end{aligned}$$

We can rewrite this system of linear equations as

$$RDf_1(A) \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Hence $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \in \ker(RDf_1(A))$.

Let $r = \text{rank}(A)$, then $\exists v_1, v_2, v_3, \dots, v_{n-r} \in \ker(A)$ by the rank-nullity theorem, and thus $v_1, v_2, v_3, \dots, v_{n-r} \in \ker(RDf_1(A))$.

Therefore $x \in \ker(RDf_1(A))$.

Note that $x \notin \text{Linspan}(v_1, v_2, v_3, \dots, v_{n-r})$.

To show this by way of contradiction assume that $x \in \text{Linspan}(v_1, v_2, v_3, \dots, v_{n-r})$.

Then $x = \sum_{i=1}^{n-r} b_i v_i$ and thus

$$Ax = A\left[\sum_{i=1}^{n-r} b_i v_i\right] = \sum_{i=1}^{n-r} b_i Av_i = 0.$$

This is a contradiction as $Ax = u$.

As $v_1, v_2, \dots, v_{n-r}$ are linearly independent and as x is not in their linear span, we see that

$$\dim \ker(RDf_1(A)) \geq n - r + 1. \quad (2)$$

We observe that $RDF_1(A)$ as a linear transformation $RDF_1(A) : \mathbb{F}^n \rightarrow \mathbb{F}^{n-1}$

Then by the rank-nullity theorem

$$\begin{aligned} \text{rank}(RDf_1(A)) &= n - \dim(\ker(RDf_1(A))) \\ &\leq n - (n - r + 1) \text{ using (2)} \\ &\leq r - 1 < r = \text{rank}(A). \end{aligned}$$

We know that

$$\text{rank}(RDf_1(A)) \in \{ \text{rank}(A), \text{rank}(A) - 1 \}$$

from Lemma 1, and thus it follows that $\text{rank}(RDf_1(A)) = r - 1$.

Conversely assume that $\text{rank}(RDf_1(A)) = \text{rank}(A) - 1$. Let $\text{rank}(A) = r$, then we know by the definition of row difference matrix of A that $\exists x \in \mathbb{F}^n$ that $RDF_1(A)x = 0$, but $Ax \neq 0$.

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

We know that $RDF_1(A)x = 0$, so

$$\begin{aligned} 0 &= \sum_{i=1}^n x_i(a_{2i} - a_{1i}) \\ 0 &= \sum_{i=1}^n x_i(a_{3i} - a_{1i}) \\ &\vdots \\ 0 &= \sum_{i=1}^n x_i(a_{ni} - a_{1i}) \end{aligned}$$

Then

$$\begin{aligned} \sum_{i=1}^n x_i a_{2i} &= \sum_{i=1}^n x_i a_{1i} \\ \sum_{i=1}^n x_i a_{3i} &= \sum_{i=1}^n x_i a_{1i} \\ &\vdots \\ \sum_{i=1}^n x_i a_{ni} &= \sum_{i=1}^n x_i a_{1i} \end{aligned}$$

$$\text{Let } \lambda = \sum_{i=1}^n x_i a_{1i}. \text{ Then } A = \begin{bmatrix} \sum_{i=1}^n x_i a_{1i} \\ \sum_{i=1}^n x_i a_{2i} \\ \vdots \\ \sum_{i=1}^n x_i a_{ni} \end{bmatrix} =$$

$$\begin{bmatrix} \lambda \\ \lambda \\ \vdots \\ \lambda \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Hence $Ax = \lambda u$.

Thus A is weakly stochastic. This completes the proof.

Lemma 2: Let A be an $n \times n$ matrix and let $u = (1, 1, \dots, 1) \in \mathbb{F}^n$. If $\exists v \in \mathbb{F}^n$ with $Av = 0$, $\langle v, u \rangle \neq 0$, then $u \notin \text{rowspan}(A)$ and hence A is not weakly stochastic.

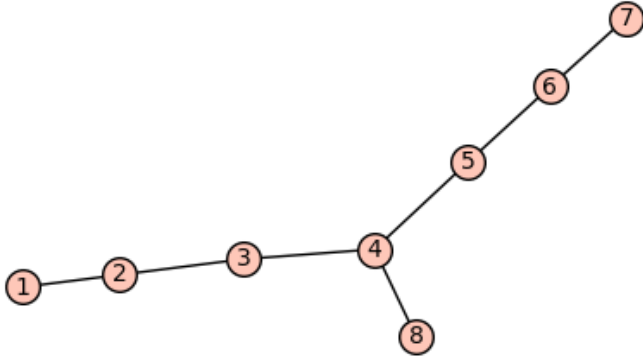
Assume by way of contradiction that $u \in \text{rowspan}(A)$, then $u = \sum_{i=1}^n c_i r_i$, where r_i are the rows of A .

$$0 \neq \langle v, u \rangle = \langle v, \sum_{i=1}^n c_i r_i \rangle = \sum_{i=1}^n c_i \langle v, r_i \rangle = \sum_{i=1}^n c_i \cdot 0 = 0$$

because from $Av = 0$ it follows that $\langle v, r_j \rangle = 0$. Thus $u \notin \text{rowspan}(A)$.

Example 2: Let A be the adjacency matrix of the graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 8\}, \{4, 5\}, \{5, 6\}, \{6, 7\}\}$.

Fig. 1. The graph A



Then the matrix for A is

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We can find a vector v here, where $v = [1, 0, -1, 0, 0, 0, 0, 1]^T$.

Given

$u = [1, 1, 1, 1, 1, 1, 1, 1]^T$, then $\langle u, v \rangle = 1$ and $Av = 0$.

Hence A is not weakly stochastic.

II. DISTANCE MATRICES

In this final section it is proved that the distance matrix of a tree is a weakly stochastic matrix.

A. Definitions

For a simple connected graph G with set of vertices $V(G) = \{v_1, v_2, \dots, v_n\}$, let d_{ij} denote the length of the shortest path between the vertices v_i and v_j . Then we define the *distance matrix of G* , as the $n \times n$ matrix with $(i, j)^{th}$ entry equal to d_{ij} . [4]–[6]

B. Trees

Assume that in this section matrices are taken over a field $\mathbb{F}$ of characteristic zero. In particular we may consider the field of rational numbers.

Theorem 2: The distance matrix $DM(T)$ of any tree T over any field is weakly stochastic.

The proof is easy. Let A be the distance matrix of a tree T , then by Graham and Pollack's theorem for distance matrices of trees [6]–[8], the determinant of the distance matrix A is given by

$$\det(A) = (-1)^{n-1} (n-1) 2^{n-2} \text{ for } n \geq 1 \quad (3)$$

We know that $\det(A) \neq 0 \iff \text{rank}(A) = n$,

and we can see by (3) that $\text{rank}(A)$ is always maximal for all $n \geq 2$. If $\text{rank}(A)$ is maximal then $RDf_1(A)$ always is

equal to $\text{rank}(A) - 1$ as $RDf_1(A)$ is an $(n-1) \times n$ matrix. Thus $\text{rank}(RDf_1(A)) \leq n-1$, but we know by Lemma 1 that $\text{rank}(RDf_1(A)) \in \{\text{rank}(A), \text{rank}(A) - 1\}$, thus $\text{rank}(RDf_1(A)) = \text{rank}(A) - 1$. Hence by theorem 1 the matrix A is weakly stochastic.

Alternatively from $\det(DM(T)) \neq 0$ we see that $DM(T)$ is an isomorphism $\mathbb{F}^n \rightarrow \mathbb{F}^n$, and hence the vector $u = (1, 1, \dots, 1)$ is in the image of $DM(T)$.

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